

Processes Governing the Ablation of Intercepted Snow

Alex C. Cebulski¹, John W. Pomeroy¹

¹Centre for Hydrology, University of Saskatchewan, Canmore, Canada,

Key Points:

- Unloading was found to be strongly associated with canopy snow load, wind shear stress, and canopy snowmelt.
- Implementation of a new model that includes both melt and dry snow unloading processes improved canopy snow load predictions.
- The fraction of snowfall sublimated in the canopy differed by up to a factor of three across existing models.

Corresponding author: Alex C. Cebulski, alex.cebulski@usask.ca

Abstract

Interception and ablation of snow in forest canopies significantly influence the quantity, timing, and phase of precipitation that reaches the ground in cold regions forests. Yet current modelling approaches have uncertain transferability across differing climate and forest types. Here, in-situ observations from a needleleaf forest in the Canadian Rockies are utilised to evaluate the theories underpinning existing canopy snow ablation models and develop novel understanding to support the development of a new canopy snow ablation model. The observations revealed that canopy snow load, wind shear stress, and canopy snowmelt were strongly associated with unloading; however, air temperature and sublimation were not. A new canopy snow ablation model was developed based on these associations and their impact on the canopy snow energy and mass balance. This model demonstrated improved performance in simulating canopy snow load relative to previous approaches, especially for melt- and wind-dominated ablation events. The improved performance in representing canopy snow load compared to existing models results from including a comprehensive representation of the mass and energy balance of intercepted snow. In contrast, all existing canopy snow models were found to omit key processes which limited accuracy of simulated snow load and partitioning of intercepted snow. The inclusion of updated melt and dry snow unloading calculations in the new model also led to more consistent partitioning of snowfall to the atmosphere versus the ground compared to existing approaches across a wide range of meteorologies.

1 Introduction

The seasonal snowpack is an essential component of global water resources (Immerzeel et al., 2020; Viviroli et al., 2020) and is increasingly threatened by rapid climate and land cover change (Aubry-Wake & Pomeroy, 2023; Fang & Pomeroy, 2023; López-Moreno et al., 2014; Szczypta et al., 2015). Vegetation cover can significantly alter the quantity, phase, and timing (Ellis et al., 2010; Essery et al., 2003; Mazzotti et al., 2021; Pomeroy & Gray, 1995; Rojas-Heredia et al., 2024; Safa et al., 2021; Sanmiguel-Valladolid et al., 2020) of snowfall reaching the ground. Snow falls in forests over half of the Northern Hemisphere (Kim et al., 2017) and over 23% of the global landmass (Deschamps-Berger et al., 2025), making forest-snow interactions crucial to understand for informed ecological, land management, and water resource decision-making. Needleleaf canopies are particularly effective at intercepting snowfall (Cebulski & Pomeroy, 2025b; Hedstrom & Pomeroy, 1998; Pomeroy & Schmidt, 1993; Storck et al., 2002), where intercepted snow is subject to increased energy inputs relative to the subcanopy snowpacks, leading to increased rates of melt, unloading, and/or sublimation (Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b; Roesch et al., 2001; Storck et al., 2002). The partitioning of snow to the atmosphere via sublimation (Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b) or to the ground through unloading and meltwater drip (Lumbrazo et al., 2022; Roesch et al., 2001; Storck et al., 2002) is highly sensitive to meteorological conditions and forest structure contributing to differing process emergence and substantial variability in subcanopy

snow accumulation. Coastal humid environments typically exhibit small sublimation losses and a larger influence of unloading and melt (Floyd, 2012; Storck et al., 2002). Here, enhanced canopy energy inputs combined with high humidity increases both solid snow unloading and melt of snow intercepted in the canopy (Lumbrazo et al., 2022; Lundquist et al., 2021; Roesch et al., 2001). High liquid water contents during rapid melt can lubricate the snow attachment to the canopy and weaken cohesive bonds, inducing unloading, much as for wet snow avalanches (Baggi & Schweizer, 2008). Conversely, the colder and drier winters in continental climates typical of the boreal forests can induce substantial canopy sublimation losses (e.g., 25–45% of annual snowfall in Essery et al., 2003) in addition to unloading (Essery et al., 2003; Gelfan et al., 2004; Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b, 2002). As a result, reliable models of snow accumulation and streamflow in forested basins rely on a comprehensive understanding of interception processes (Clark et al., 2015; Essery et al., 2003; Pomeroy et al., 1998a; Verseghy, 2017; Wheeler et al., 2022).

Canopy snow models have demonstrated variable transferability across different climates and forest types (Essery et al., 2003; Gelfan et al., 2004; Lumbrazo et al., 2022; Lundquist et al., 2021), which has been attributed to limiting the performance in predicting forest snowpacks in hydrological model intercomparisons (Krinner et al., 2018; Rutter et al., 2009). Recent studies have emphasized the importance of distinguishing between initial snow interception and subsequent ablation processes (Cebulski & Pomeroy, 2025a, 2025b). This separation allows for individual parameterisations for distinct processes, improving both process representation and the modular design of contemporary models, thereby supporting broader applicability across diverse environments and model structures (Clark et al., 2015; Pomeroy et al., 2022). In addition, Lundquist et al. (2021) and Cebulski & Pomeroy (2025b) demonstrated that canopy snow processes can be more accurately represented without the concept of a maximum canopy snow load, which is included in initial accumulation parameterisations (Andreadis et al., 2009; Hedstrom & Pomeroy, 1998). Moreover, existing canopy snow accumulation parameterisations often incorporate ablation routines, likely as a result of some ablation included in measurements of interception efficiency (Cebulski & Pomeroy, 2025a; Hedstrom & Pomeroy, 1998; Katsushima et al., 2023; Schmidt & Gluns, 1991). For example, Hedstrom & Pomeroy (1998) found that interception efficiency declined as the canopy is loaded with snow, while Storck et al. (2002) found a constant interception efficiency of 0.6 and is also typically combined with a maximum canopy snow load. Consequently, both the Hedstrom & Pomeroy (1998) and Storck et al. (2002) parameterisations have significantly lower interception efficiencies—prior to canopy snow ablation—compared to the observations by Cebulski

& Pomeroy (2025b). Staines & Pomeroy (2023) and Cebulski & Pomeroy (2025b) show that interception efficiency is best predicted by canopy density alone—consistent with some rainfall interception parameterisations (e.g., Valante et al., 1997; Zhong et al., 2022)—and that snow load has a small influence on canopy density, challenging the Hedstrom & Pomeroy (1998) and Storck et al. (2002) initial snow interception theories. Together these studies (Cebulski & Pomeroy, 2025a, 2025b; Lundquist et al., 2021) emphasize the need to revisit canopy snow ablation parameterisations.

Unloading of intercepted snow due to the ground has previously been shown to be associated air temperature (Katsushima et al., 2023; Roesch et al., 2001; Schmidt & Pomeroy, 1990), ice-bulb temperature—which more accurately represents the cooling effect of sublimation compared to air temperature—(Ellis et al., 2010; Floyd, 2012), canopy snowmelt rate (Storck et al., 2002), and wind speed (see Supporting Information for example relationships). Each of these factors were also found to be dependent on canopy snow load and time. While Hedstrom & Pomeroy (1998) did not make direct observations of canopy snow unloading, they proposed a parameterisation based on canopy snow load and time. In addition to the empirical observations by the aforementioned studies, physical reasoning also supports the inclusion of these processes. For example, melt promotes unloading through loss of structural integrity, particle bond weakening, and lubrication of intercepted snow. Wind drag promotes unloading through shear stress applied to intercepted snow, wind erosion through direct entrainment in the atmosphere of intercepted snow, and branch movement. Branch elasticity increases with temperature (Schmidt & Pomeroy, 1990) which can increase the likelihood of unloading due to increasing branch angle under a snow load and decreased stiffness to resist swaying in a turbulent wind.

Melt of snow intercepted in the canopy is typically represented by either an energy balance approach (Andreadis et al., 2009; Clark et al., 2015; Essery et al., 2025; e.g., Parviainen & Pomeroy, 2000), as a function of air temperature (Roesch et al., 2001), or a function of ice-bulb temperature (Ellis et al., 2010; Floyd, 2012). Sublimation is generally represented using a coupled energy and mass balance approach (e.g., Essery et al., 2003; Pomeroy et al., 1998b; Verseghy, 2017). The Essery et al. (2003) and Pomeroy et al. (1998b) approaches differ in that Pomeroy et al. (1998b) does not include longwave radiation in the canopy snow energy balance. Both the Essery et al. (2003) and Pomeroy et al. (1998b) parameterisations decrease the latent heat flux from snow intercepted in the canopy as the canopy fills up with snow and its specific surface area decreases. However, these parameterisations are based on estimates of maximum canopy snow load which may underestimate true maxima

(Cebulski & Pomeroy, 2025b; Lundquist et al., 2021; Storck et al., 2002) and should be reconsidered using a larger maximum load or an approach that avoids prescribing a maximum load. The merits of including more physically based energy balance methods compared to more empirically based functions for calculating snowmelt and sublimation have not been directly assessed using an event-based process investigation. Some models vertically discretize the canopy into multiple layers (e.g., Clark et al., 2015) or into two layers (e.g., Essery et al., 2025) to better capture vertical gradients in the canopy energy balance—for instance, enhanced solar radiation and turbulent exchanges near the canopy top relative to the lower canopy. However, uncertainties in generating forcing data at different canopy heights can limit the effectiveness of these multi-layer approaches. In contrast, other schemes represent the canopy as a single layer (Pomeroy et al., 2022; Verseghy, 2017).

Quantifying individual canopy snow ablation processes including unloading, wind redistribution, melt, drip, and sublimation remains challenging, even with sophisticated lysimetry (Storck et al., 2002) and eddy-covariance systems (Conway et al., 2018; Harding & Pomeroy, 1996; Harvey et al., 2025). Consequently, some canopy snow ablation parameterisations have been developed using methods, such as above canopy albedo, which do not distinguish individual processes (Bartlett & Verseghy, 2015; Roesch et al., 2001). While these approaches offer useful indices of model performance, they provide limited insight into the accuracy of individual process representations. Lysimeter-based measurements offer more direct process-level observations, but interpretation of their observations is made uncertain by freeze-thaw cycles and concurrent processes (Floyd, 2012; MacDonald, 2010; Storck et al., 2002). A hybrid diagnostic approach that combines individual process measurements with simulations and employs observations such as canopy snow load from a weighed tree has yet to be applied but is explored in this study.

The objective of this study is to better understand and predict the influences of meteorology and snow load on intercepted snow ablation. This study specifically looks at canopy snow ablation after snowfall and initial interception. Cebulski & Pomeroy (2025b) has addressed processes governing the initial accumulation of snow in the canopy.

The specific research questions this research aims to address include:

1. How do snow load, air temperature, humidity, wind exposure, canopy snow sublimation, and snowmelt influence the rate of canopy snow unloading?

2. To what extent do current theoretical models of canopy snow ablation align with detailed in-situ observations?
3. What modifications to existing models, are necessary to accurately represent ablation of snow intercepted in the canopy and what is the improvement in performance from these modifications?

2 Methods

2.1 Study Site

The observations presented in this study were collected at Fortress Mountain Research Basin (FMRB), Alberta, Canada, 51° N, 115° W, a continental headwater basin in the Canadian Rockies. The site is located 2100 m above sea level on a ridge covered by mature fir and spruce forest. Air temperature, humidity, and wind speed were measured at a height of 4.3 m at Forest Tower Station (Figure 1). Shear stress was calculated using the EddyPro software (LI-COR Biosciences) based on high-frequency wind measurements from a CSAT3 three-dimensional sonic anemometer (Campbell Scientific) installed at 3.0 m at Forest Tower station. The CSAT3 was occasionally covered in snow during the analysis period, and thus to provide a complete record of shear stress, a linear relationship was established between shear stress derived from the CSAT3 and the square of wind speed measured at 4.3 m at the Forest Tower station ($R^2 = 0.71$, $p < 0.05$). This relationship was then used to gap-fill shear stress during periods when the CSAT3 was snow-covered. The precipitation rate was measured by an Alter-shielded OTT Pluvio weighing precipitation gauge installed 2.6 m above ground at the adjacent Powerline Station (Figure 1). Incoming and outgoing solar radiation was measured by a Kipp & Zonen CNR4 4-Component Net Radiometer installed 3.27 m above the ground at Fortress Ridge Station 2.0 km to the northwest of Forest Tower station (50.8° N, 115.2° W). This windy exposed site was selected to reduce snow accumulation on the radiometers, in addition to the CNR4's built-in heating element. Three subcanopy lysimeters, consisting of plastic horse-watering troughs with an opening of 0.9 m² and depth of 20 cm suspended from a load cell, were installed to measure subcanopy snow accumulation. A weighed tree lysimeter (subalpine fir) suspended from a load cell (Artech S-Type 20210-100) measured the weight of canopy snow load (kg) and was scaled to an areal estimate of snow load (mm) using snow surveys as in Pomeroy & Schmidt (1993). To isolate the subcanopy lysimeter measurements of canopy snow unloading from throughfall, 15-min intervals were selected that had no atmospheric precipitation based on the precipitation gauge at Powerline. The weighed tree and timelapse imagery also were used to confirm there was no atmospheric precipitation and to identify periods where

snow was intercepted in the canopy and ablation was possible. Four tipping bucket rain gauges—3 Texas Electronics TR-525M and 1 Hyquest Solutions TB4MM—were installed along a 15 m transect adjacent to the dense canopy subcanopy lysimeter to quantify sub-canopy drip from melting intercepted snow. Additional details on this study site and the meteorological and lysimeter measurements have been described in Cebulski & Pomeroy (2025b).

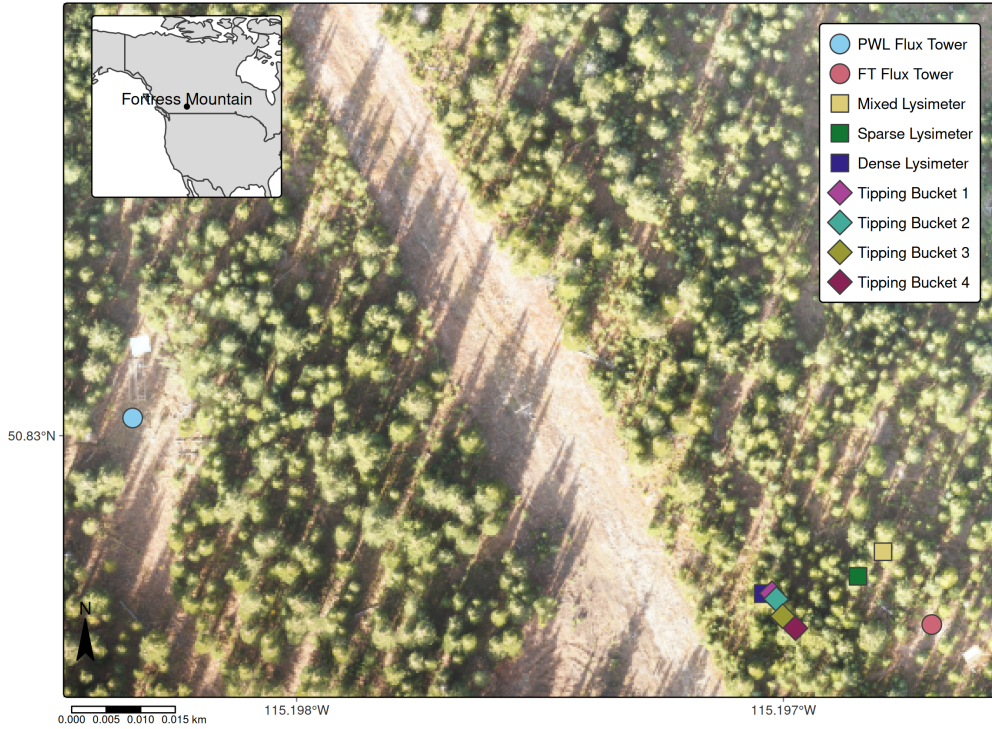


Figure 1: Map showing the location of flux towers and lysimeter instruments. The inset map on the upper left shows the regional location of Fortress Mountain Research basin. Flux towers are denoted by a circle, lysimeters by a square, and tipping bucket rain gauges by diamonds.

2.2 The Cold Regions Hydrological Model Platform

The Cold Regions Hydrological Model Platform (CRHM) was used to implement calculations of the canopy snow energy and mass budget. A full description of the CRHM platform is described in Pomeroy et al. (2022) and the source code is available at <https://github.com/srlabUsask/crhmcode>. The climate forcing data used to run CRHM was provided from station-based fifteen-minute interval measurements of air temperature, relative humidity, wind speed, precipitation, and incoming solar radiation from the station measurements. CRHM incorporates a flexible modular

design allowing the user to select various parameterisations that represent hydrological processes. The phase of atmospheric precipitation was determined from the energy balance of falling hydrometeors (Harder & Pomeroy, 2013). A new CRHM module was created here to simulate the coupled mass and energy balance of snow intercepted in the canopy. The energy balance is described in detail in Section 2.5 and the updates to the mass balance through revisions to the canopy snow unloading empirical functions are presented in Section 3.1.1 and Section 3.1.2. Due to uncertainties in estimating model forcing data and measuring canopy snow ablation at varying canopy heights, a single layer canopy snow model was implemented in CRHM.

2.3 Canopy Snow Mass Balance

The following mass balance as described in Cebulski & Pomeroy (2025a) was implemented in CRHM to track the state of canopy snow load (L , mm) over time:

$$\frac{dL}{dt} = [q_{sf} - q_{tf} + q_{ros}] - [q_{unld}^{melt} + q_{unld}^{dry}] - q_{drip} - q_{wind}^{veg} - q_{sub}^{veg} \quad (1)$$

where $\frac{dL}{dt}$ is the change in canopy snow load over time, (mm s^{-1}), q_{sf} is the snowfall rate (mm s^{-1}), q_{tf} is the throughfall rate (mm s^{-1}), q_{ros} is the rate of rainfall falling on snow intercepted in the canopy (mm s^{-1}), q_{unld}^{melt} is the unloading rate due to melt (mm s^{-1}), q_{unld}^{dry} is the dry snow unloading rate due to shear stress on snow, wind erosion, branch movement, structural degradation, bond weakening and increased elasticity of branches and other non-melt related processes (mm s^{-1}), q_{drip} is the canopy snow drip rate (mm s^{-1}) resulting from canopy snowmelt (q_{melt} , mm s^{-1}), q_{wind}^{veg} is the wind transport rate in or out of the control volume (mm s^{-1}), and q_{sub}^{veg} is the intercepted snow sublimation rate (mm s^{-1}) which also includes evaporation of liquid water in the canopy. Figure 1 in Cebulski & Pomeroy (2025a) presents a visual representation of this mass balance.

2.4 Mass Balance Parameterisations

In addition to the updated canopy snow model presented in this study, hereafter referred to as CP25 (described in Section 3.1.1 and Section 3.1.2), three other canopy snow models were implemented in CRHM following previous studies by Ellis et al. (2010) and Floyd (2012) (E10), Roesch et al. (2001) (R01), and Andreadis et al. (2009) who built on observations by Storck et al. (2002) (SA09) (Table 1). The E10 model includes canopy snow sublimation as described in Pomeroy et al. (1998b), dry snow unloading as a function of canopy snow load from Hedstrom & Pomeroy (1998) with modifications described in Ellis et al. (2010) and Floyd (2012) to handle canopy snow melt and drip processes using an ice-bulb temperature threshold. The R01

model represents dry snow unloading, melt, and drip using linear functions of wind speed and air temperature, while sublimation is simulated using the parameterisation from Pomeroy et al. (1998b). The SA09 model unloads snow as a ratio of the canopy snowmelt rate, following observations by Storck et al. (2002) and does not include dry snow unloading. The snowmelt rate from Equation 3 was used to calculate the unloading rate for the Andreadis et al. (2009) unloading and thus differs from the energy balance routine described in their study. Canopy snow sublimation in SA09 was represented based on the Essery et al. (2003) parameterisation (Equation 10) as implemented in CP25.

Table 1: Description of canopy snow unloading parameterisations.

Name	Study	Dry Snow Unloading (q_{und}^{dry})	Melt Induced Unloading (q_{und}^{melt})
CP25	This study	$f(\text{snow load, wind shear stress})$, Eq. 12	$f(\text{snow load, canopy snowmelt rate})$, Eq. 14
E10	Ellis et al., (2010); Hedstrom & Pomeroy (1998)	$f(\text{snow load})$, Eq. 5 in Hedstrom & Pomeroy (1998)	$f(\text{ice-bulb temp.})$, see description following Eq. 28 in Cebulski & Pomeroy (2025a)
SA09	Storck et al., (2002); Andreadis et al., (2009)	NA	$f(\text{canopy snowmelt rate})$, Eq. 33 in Andreadis et al., (2009)
R01	Roesch et al., (2001)	$f(\text{snow load, wind speed})$, Eq. 15 in Roesch et al., (2001)	$f(\text{snow load, air temp.})$, Eq. 14 in Roesch et al., (2001)

The retention of canopy snow meltwater differs between the four canopy snow models implemented in CRHM. For E10, liquid meltwater is not retained in the canopy and immediately drips before it can evaporate. The R01 model also does not represent evaporation of liquid meltwater as canopy snow is not explicitly separated between solid snow unloading and melt/drip. The canopy liquid water storage capacity (L_{max}^{liq} , mm) from SA09 and CP25 was calculated as:

$$L_{max}^{liq} = Lh + nLAI \quad (2)$$

where h is the liquid meltwater holding capacity that canopy can retain against gravity and n is a storage constant of the vegetation elements. Andreadis et al. (2009) estimated h as 0.035 (-) and n as e^{-4} . In this study, h was set to 0.01 (-), and n was defined as $C_c \cdot 0.2$. A one-sided effective plant (leaf) area index (LAI) was used, including the clumping factor to account for needleleaf canopy structures, following the approach of Ellis et al. (2010) for rainfall interception.

2.5 Canopy Snow Energy Balance

The CP25 and SA09 canopy snow models implemented the following energy balance approach to calculate the energy available for melting snow intercepted in the canopy (Q_{melt} , W m^{-2}) and to track the canopy snow temperature over time ($\frac{\Delta T_{vs}}{\Delta t}$, K s^{-1}). The energy balance is expressed as:

$$Q_{melt} = Q_{sw} + Q_{lw} + Q_p + Q_h + Q_l - [C_p^{ice} L \frac{\Delta T_{vs}}{\Delta t}] \quad (3)$$

where Q_{sw} and Q_{lw} (W m^{-2}) are the net shortwave and longwave radiation heat fluxes to the canopy snow, Q_p (W m^{-2}), is the advective energy rate, and Q_l and Q_h (W m^{-2}), are the turbulent fluxes of latent heat and sensible heat respectively (positive towards canopy snow), and C_p^{ice} ($\text{J kg}^{-1} \text{K}^{-1}$) is the specific heat capacity of ice. Figure 2 in Cebulski & Pomeroy (2025a) shows a visual representation of this energy balance.

2.6 Energy Balance Parameterisations

The E10 and R01 parameterisations relied on physically guided relationships to simulate sublimation and melt and are described in full detail in their respective articles and are also summarised in Cebulski & Pomeroy (2025a). The following section describes the energy balance parameterisations implemented in the CP25 and SA09 models.

Q_{sw} was determined as:

$$Q_{sw} = Q_{sw}^{in} \cdot (1 - \alpha_s) \cdot \tau_{50}^{veg} \quad (4)$$

where Q_{sw}^{in} is the downwelling shortwave radiation (W m^{-2}), α_s is the albedo of snow intercepted in the canopy (-), τ_{50}^{veg} (-) is the canopy transmittance to Q_{sw}^{in} . Equation 10 from Pomeroy et al. (2009) was used to determine τ_{50}^{veg} , using half of the leaf area index (LAI) based on studies Weiskittel et al. (2009) and Kesselring et al. (2024) that approximately 50% of the total leaf area is concentrated in the upper half of coniferous canopies. Q_{sw} provides an approximation of the net shortwave radiation

to all snow intercepted in the canopy and is a simplification from using a partial differential equation to determine the radiation incident to individual height layers within the canopy. Upwelling shortwave radiation reflected off the surface snowpack is considered negligible contribution to the snow intercepted in the canopy as it is primarily blocked by vegetation elements underlying the canopy snow (Pomeroy et al., 2009).

Q_{lw} was approximated as:

$$Q_{lw} = \downarrow Q_{lw}^{atm} + \uparrow Q_{lw}^{veg} - Q_{lw}^{vs} \quad (5)$$

where Q_{lw}^{atm} is the downwelling longwave radiation from the atmosphere (W m^{-2}), Q_{lw}^{veg} is the longwave radiation upwelling from vegetation elements underlying snow intercepted in the canopy (W m^{-2}), and Q_{lw}^{vs} (W m^{-2}) is the outgoing longwave radiation from the top and bottom of the canopy snow layer calculated as:

$$Q_{lw}^{vs} = 2\epsilon_s \sigma T_{vs}^4 \quad (6)$$

where ϵ_s is the emissivity (-) of snow taken as 0.99 and σ is the Stefan-Boltzmann ($5.67 \times 10^{-10} \text{ W m}^{-2} \text{ K}^{-4}$). Q_{lw}^{atm} was approximated in this study as in Sicart et al. (2006) to represent the influence of atmospheric moisture and clouds on emissivity. Q_{lw}^{veg} was calculated with the assumption that canopy elements are in equilibrium with the air temperature plus any increase in vegetation temperature from the extinction of Q_{sw}^{in} in the canopy (Pomeroy et al., 2009, Eq. 4).

Q_p was calculated as:

$$Q_p = [C_p^{liq} m_r (T_r - T_{vs}) + C_p^{ice} m_s (T_s - T_{vs})] / \Delta t \quad (7)$$

where C_p^{liq} is the specific heat capacity of liquid water ($\text{J kg}^{-1} \text{ K}^{-1}$), m_r is the specific mass of liquid water in precipitation (mm), T_r is the rainfall temperature (K), m_s is the specific mass of snow in precipitation (mm), and T_s is the snowfall temperature (K).

Q_h was calculated as:

$$Q_h = \frac{\rho_a}{r_a} C_p^{air} (T_a - T_{vs}) \quad (8)$$

where ρ_a is the air density (kg m^{-3}), C_p^{air} is the specific heat capacity of air ($\text{J kg}^{-1} \text{K}^{-1}$), T_a is the air temperature, and r_a is the aerodynamic resistance (s m^{-1}) which was approximated from Equation 4 in Allan et al. (1998) as:

$$r_a = \frac{\log\left(\frac{z_T - d_0}{z_0}\right) \log\left(\frac{z_u - d_0}{z_0}\right)}{\kappa^2 u_z} \quad (9)$$

where z_T is the height of temperature measurement (m), d_0 is the displacement height (m) which was approximated as $2/3^{\text{rd}}$ the mean canopy height, z_0 is the roughness length (m) which was approximated as $1/10^{\text{th}}$ of the mean canopy height, z_u is the wind speed measurement height (m), κ is von Kármán's constant, 0.41 (-), and u_z is the wind speed measurement at z_u (m s^{-1}).

Q_l was calculated as:

$$Q_l = \frac{\rho_a}{r_i + r_a} (q_a(T_a) - q_{vs}(T_{vs})) \quad (10)$$

where r_i is a resistance for transport of moisture from intercepted snow to the canopy air space (Eq. 28 in Essery et al., 2003), $q_a(T_a)$ and $q_{vs}(T_{vs})$ are the specific humidity (-) at the air temperature and canopy snow temperature, respectively. r_i was calculated following the concept of how full the canopy is with snow as introduced by Pomeroy & Schmidt (1993) with modifications to incorporate a larger maximum canopy snow load capacity of 50 mm based on observations by Storck et al. (2002), Floyd (2012), and Cebulski & Pomeroy (2025b).

The above sensible and latent heat flux equations assume neutral atmospheric stability conditions, which is supported by the uncertainty of stability correction in forest canopies (Conway et al., 2018) and mountain environments in winter (Helgason & Pomeroy, 2012a). Solving Equation 3 requires an iterative solution to determine ΔT_{vs} and the remaining terms which are also a function of T_{vs} .

2.7 Influence of Predictive Variables on Unloading

The following hypotheses were tested to determine dominant predictors of unloading:

- a. Melt promotes unloading through loss of structural integrity, particle bond weakening, and lubrication of intercepted snow.
- b. Sublimation promotes unloading via structural degradation and bond weakening of intercepted snow.

- c. Wind drag promotes unloading through shear stress applied to intercepted snow, wind erosion through direct entrainment in the atmosphere of intercepted snow, and branch movement.
- d. Increasing air temperature promotes unloading by increasing the elasticity of branches and its association with melt and/or sublimation.

Distributions of air temperature, wind speed, relative humidity, canopy snow unloading, melt, and sublimation were compared between periods with simulated canopy snow melt and periods without melt. A canopy snowmelt threshold of 0 mm hr^{-1} was used to define non-melt periods, with values greater than 0 mm hr^{-1} indicating melt periods. These comparisons were used to evaluate whether the samples represent the same underlying distribution, thereby determining whether the hypotheses and statistical relationships should be analysed separately for melt and non-melt periods. A two sample non-parametric Wilcoxon signed-rank test was used to evaluate whether distributions of the variables between the two periods differ by a location shift of zero. In addition, the non-parametric Kolmogorov-Smirnov test was applied to compare the two samples as a whole and assess whether they were drawn from the same continuous distribution.

The subcanopy lysimeter unloading measurements had a high relative instrument error due to the relatively small accumulation of unloaded snow over the 15-min intervals. To improve instrument accuracy, whilst maintaining consistency of the unloading measurements with the independent variables—and minimize issues related to with temporal autocorrelation—the 15-min interval measurements of unloading were aggregated into bins defined by ranges of the predictor variables. Independent variable bins that had less than 0.1 mm of accumulated snow were removed, resulting in a mean instrument error of $\pm 2\%$ for the remaining bins. Air temperature and wind speed were measured at Forest Tower station, canopy snow load from the weighed tree lysimeter (scaled to the canopy of each respective subcanopy lysimeter), and canopy snowmelt and sublimation simulated using CRHM with the CP25 model as described in the previous section.

Relationships representing the canopy snow unloading process using differing combinations of predictor variables including canopy snow load, air temperature, ice-bulb temperature, ice-bulb temperature depression (air temperature subtracted by ice-bulb temperature), wind speed, shear stress, canopy snowmelt were evaluated using the mean bias, root-mean-square error (RMSE), coefficient of determination (R^2), and coefficient of agreement (Willmott, 1982). Linear relationships that did not include an intercept term had R^2 values adjusted following (Kozak & Kozak, 1995). Binning

unloading across the different predictor variables produced distinct unloading distributions for each predictor, which precluded the use of Akaike’s ‘An Information Criterion’ metric for evaluating model performance. Linear relationships were fit using ordinary least squares regression, and nonlinear relationships using nonlinear least squares regression. Least squares regression was used because it provides a transparent and physically interpretable fit for the hypothesized process-based relationships. Regression model assumptions—including normality and homoscedasticity—were assessed using Shapiro-Wilk and Breusch-Pagan tests respectively. Multicollinearity was evaluated via Variance Inflation Factors, Pearson’s correlation, and non-linear correlations amongst predictors were assessed using Spearman’s rank correlation coefficient. Relationships failing the homoscedasticity test (Breusch-Pagan $p < 0.05$) were modelled using generalized least squares (GLS) regression, which accounts for unequal residual variances by appropriately weighting observations. P-values are not reported for GLS coefficients, as heteroscedasticity violates the t-distribution assumptions underlying standard significance tests.

2.7.1 Melt Driven Unloading

Due to the physical interdependence of the canopy snowmelt rate on canopy snow load, melt-driven relationships were evaluated using a dimensionless snowmelt rate (s^{-1}) calculated as the snowmelt rate ($mm\ s^{-1}$) divided by snow load (mm). A mass balance approach was also incorporated to determine the unloading rate resulting from canopy snowmelt (q_{unld}^{melt} , $mm\ s^{-1}$). The effect of the canopy snowmelt rate (q_{melt}) on unloading was then assessed by fitting a linear model using an ordinary least squares regression. Since direct measurements of canopy snow unloading are challenging to obtain independently of canopy snowmelt drainage (Storck et al., 2002), the mass balance introduced in Equation 1 was incorporated to determine q_{unld}^{melt} as a residual. During intervals without $[q_{sf} - q_{tf} + q_{ros}]$ or q_{wind}^{veg} Equation 1 was simplified and rearranged to:

$$q_{unld}^{melt} = -\frac{dL}{dt} - q_{drip} - q_{unld}^{dry} - q_{sub}^{veg} \quad (11)$$

While some components of the canopy snow mass balance can be measured directly, such as $\frac{\Delta L}{\Delta t}$ with the weighed tree, sublimation of canopy snow is more difficult to quantify directly especially in forested mountain environments (Conway et al., 2018; Harding & Pomeroy, 1996; Helgason & Pomeroy, 2012b; Parviainen & Pomeroy, 2000) and thus q_{sub}^{veg} was simulated in this study in CRHM using Equation 10 as described in Essery et al. (2003). q_{drip} was measured where possible using the rain gauges,

however problems with freezing of liquid water in the tipping bucket mechanisms limited the ability to measure q_{drip} reliably. Thus, q_{drip} was also estimated using simulations of canopy snowmelt (q_{melt}) in CRHM as in Equation 3, with storage limited by the canopy liquid water holding capacity and drainage of excess water. Canopy snow ablation periods that were dominated by melt were selected for calculating q_{unld}^{melt} where the contribution of q_{unld}^{dry} and q_{sub}^{veg} to canopy snow ablation was less than 5%.

2.8 Model Evaluation

Performance of the CP25, E10, R01, and SA09 models was assessed using the following error metrics: mean bias (MB), root-mean-squared error (RMSE), Nash-Sutcliffe Efficiency (NSE; Nash & Sutcliffe (1970)), and Kling-Gupta Efficiency (KGE; Gupta et al. (2009); Clark et al. (2021)). Observed canopy snow ablation from the weighed tree was compared against model simulations to compute these error metrics. To quantify the sampling uncertainty related to potential influence of outlier events, a bootstrap resampling procedure with 1000 replicates was applied following a similar approach to Clark et al. (2021); however, instead of using non-overlapping blocks defined by individual years, differing combinations of entire canopy snow ablation events were resampled as discrete data blocks.

3 Results

3.1 Unloading Relationships

Results from the non-parametric Wilcoxon signed-rank test indicated that air temperature, wind speed, unloading rate, sublimation rate, and snowmelt rate differed significantly between melt and non-melt periods ($p < 0.05$, Figure 2). Relative humidity did not differ significantly based on the Wilcoxon test ($p > 0.05$). The non-parametric Kolmogorov-Smirnov test indicated that all predictors originated from differing distributions ($p < 0.05$). These statistical differences, together with the distinct physical processes operating during melt and non-melt (dry snow unloading) conditions, supported treating the two event types separately as in subsequent analyses (see Supporting Information for full test statistics).

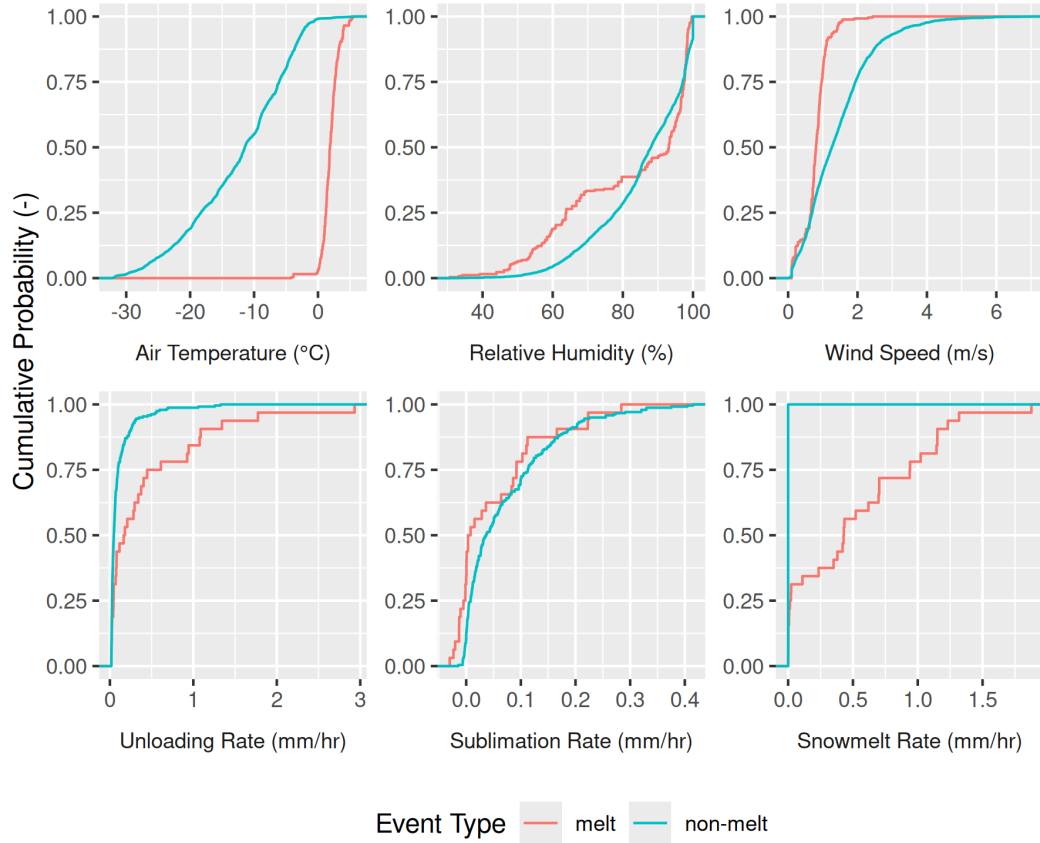


Figure 2: Comparison of distribution of air temperature, relative humidity, wind speed, unloading rate over unloading events with canopy snow melt and without. Note: Unloading data has been binned to 6 hours to ensure enough snow for satisfactory instrument error.

3.1.1 Dry Snow Unloading

A linear relationship with shear stress and an exponential relationship with wind speed, each combined with snow load, produced the lowest errors when predicting unloading during the non-melt periods. The RMSE values were 0.155 mm hr⁻¹ and 0.11 mm hr⁻¹ for the shear stress and wind speed relationships, respectively, with corresponding R^2 values of 0.58 and 0.54 (Table 2). In addition to model approximation error, uncertainty arose from instrument limitations and from unloading processes unrelated to wind. The shear stress relationship fit using ordinary least squares violated the homoscedasticity assumption (Breusch-Pagan test $p < 0.05$; see Supporting Information section), with residual variance increasing for higher unloading rates. To address this issue, generalized least squares regression was applied, which accounts for differing residual variances by appropriately weighting observations.

Additional relationships that had differing combinations of canopy snow load, shear stress, sublimation, air temperature, or ice-bulb temperature depression all produced higher RMSE values and lower R^2 and d statistics than the wind speed and shear stress relationships (Table 2). Furthermore, a model that included snow load, shear stress, and air temperature was statistically insignificant ($p > 0.05$). Several predictor combinations—specifically shear stress with wind speed, air temperature with ice-bulb temperature depression, and sublimation rate with ice-bulb temperature depression—were strongly correlated (Pearson or Spearman correlation coefficients > 0.7) and therefore were not used together in any model evaluation. The full correlation matrices are provided in the Supporting Information.

Although the wind speed relationship had reduced RMSE, the higher R^2 and d statistics and lower mean bias, coupled with the better physical representation of kinetic energy, indicate that shear stress is the preferred predictor of dry snow unloading (Table 2). Accordingly, the dry unloading rate, q_{unld}^{dry} , was represented as a linear function of shear stress:

$$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a \quad (12)$$

where τ_{mid} is the shear stress at mid-canopy height and a is a fitting constant.

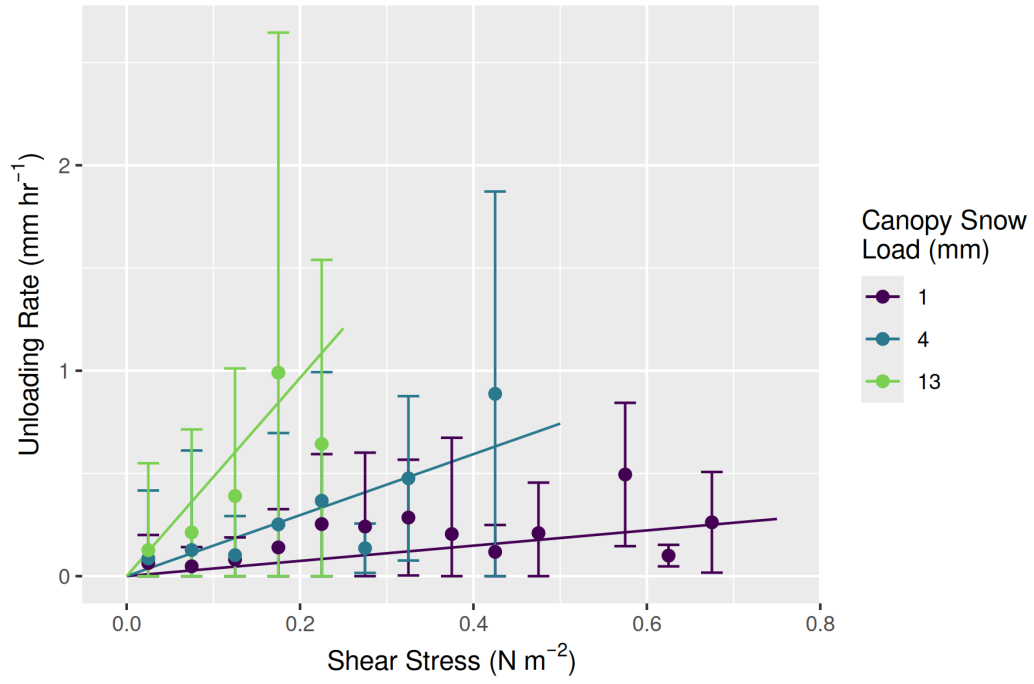


Figure 3: The canopy snow unloading rate measured by the subcanopy lysimeters versus shear stress during periods without canopy snowmelt. The dots represent mean unloading rates within bins of shear stress for three canopy snow load levels; error bars indicate ± 1 standard deviation. The fitted lines show predictions from Equation 12. The canopy snow load corresponds to the midpoint of each bin (0–2, 2–6, 6–20 mm).

Table 2: Regression statistics for dry snow unloading with various predictor combinations. Coefficients are shown for hourly unloading, derived using ordinary least squares (OLS), non-linear least squares (NLS), and generalised least squares (GLS) regression. Error metrics include mean bias (MB), root-mean-squared error (RMSE), coefficient of determination (R^2), and index of agreement (d). Row names (a) and (b) indicate coefficients, with p(a) and p(b) as their p-values. Equation variables include: dry snow unloading (q_{unld}^{dry} , mm hr⁻¹), canopy snow load (L , mm), mid canopy shear stress (τ_{mid} , N m⁻²), mid canopy wind speed (u_{mid} , m s⁻¹), canopy snow sublimation rate (q_{subl}^{veg} mm hr⁻¹), air temperature (T_a , °C), and ice-bulb temperature (T_i , °C).

	Snow Load	Shear Stress	Wind Speed	Sublimation	Shear Stress, Air Temperature	Shear Stress, Ice-Bulb dep.	Shear Stress, Sublimation
Fit	GLS	GLS	NLS	OLS	OLS	GLS	GLS
Eq	$q_{unld}^{dry} = L \cdot a$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a$	$q_{unld}^{dry} = L \cdot u_{mid} \cdot a \cdot e^{b \cdot u_{mid}}$	$q_{unld}^{dry} = L \cdot q_{subl}^{veg} \cdot a$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a + T_a \cdot b$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a + (T_a - T_i) \cdot b$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a + q_{subl}^{veg} \cdot b$
MB	0.001	0.008	0.048	0.071	0.024	0.006	-0.008
RMSE	0.159	0.155	0.11	0.157	0.199	0.255	0.19
R^2	0.24	0.58	0.54	0.16	0.23	0.31	0.21
d	0.53	0.68	0.67	0.49	0.57	0.59	0.57
a	2.28×10^{-02}	3.71×10^{-01}	4.62×10^{-03}	1.55×10^{-01}	3.00×10^{-01}	3.33×10^{-01}	2.72×10^{-01}
p(a)	NA for GLS	NA for GLS	p < 0.05	p < 0.05	p < 0.05	NA for GLS	NA for GLS
b	NA	NA	3.93×10^{-01}	NA	-3.96×10^{-03}	7.26×10^{-02}	7.09×10^{-01}
p(b)	NA	NA	p < 0.05	NA	n.s.	NA for GLS	NA for GLS

3.1.2 Melt Driven Unloading

Over the canopy snowmelt periods a linear relationship including canopy snow load (mm), simulated dimensionless canopy snowmelt rate (s^{-1}), and shear stress (N m^{-2}) had the strongest relationship with unloading and drip measured from the subcanopy lysimeters ($p < 0.05$, $R^2 = 0.49$, $\text{RMSE} = 0.355 \text{ mm hr}^{-1}$) (Table 3). A separate relationship including canopy snow load and the dimensionless canopy snowmelt rate explained less variability ($p < 0.05$, $R^2 = 0.37$) with a slightly higher RMSE of 0.39 mm hr^{-1} , suggesting wind driven unloading is also prevalent when intercepted snow is melting. In contrast, the temperature-based predictors exhibited lower performance ($p < 0.05$, $R^2 \leq 0.1$, $\text{RMSE} \approx 0.5 \text{ mm hr}^{-1}$). The superior performance of the dimensionless snowmelt rate compared with air temperature and ice-bulb temperature predictors reflects its stronger physical basis, as it represents the total energy available for melt from all components of the energy balance. Certain predictor pairs—namely shear stress with wind speed and air temperature with ice-bulb temperature—displayed strong linear and nonlinear associations (Pearson or Spearman coefficients > 0.7) and were therefore not included together in any model. Canopy snowmelt rate, air temperature, and ice-bulb temperature exhibited moderate correlation (Pearson or Spearman coefficients > 0.5) but were excluded as their known physical interdependence (full correlation statistics are provided in the Supporting Information).

Table 3: Regression statistics for melt-induced unloading with various predictor combinations using ordinary least squares (OLS). Coefficients are shown for hourly unloading. Error metrics include mean bias (MB), root-mean-squared error (RMSE), coefficient of determination (R^2), and index of agreement (d). Row names (a) and (b) indicate coefficients, with p(a) and p(b) as their p-values. Equation variables include: melt driven unloading (q_{unld}^{melt} , mm hr^{-1}) and canopy snow melt rate (q_{melt} , mm hr^{-1}), with the remaining variables defined in Table 2.

	Dimensionless Snowmelt Rate, Shear Stress	Dimensionless Snowmelt Rate	Air Temperature	Ice-Bulb Temperature
Fit	OLS	OLS	OLS	OLS
Eq	$q_{unld}^{melt} = L \cdot \frac{q_{melt}}{L} \cdot a + \tau_{mid} \cdot b$	$q_{unld}^{melt} = L \cdot \frac{q_{melt}}{L} \cdot a$	$q_{unld}^{melt} = L \cdot T_a \cdot a$	$q_{unld}^{melt} = L \cdot T_i \cdot a$
MB	0.027	0.099	0.128	0.243
RMSE	0.355	0.39	0.518	0.463
R^2	0.49	0.37	0.1	0.08
d	0.68	0.69	0.55	0.44
a	2.21×10^{-01}	2.74×10^{-01}	1.09×10^{-01}	1.42×10^{-01}

Table 3: Regression statistics for melt-induced unloading with various predictor combinations using ordinary least squares (OLS). Coefficients are shown for hourly unloading. Error metrics include mean bias (MB), root-mean-squared error (RMSE), coefficient of determination (R^2), and index of agreement (d). Row names (a) and (b) indicate coefficients, with $p(a)$ and $p(b)$ as their p -values. Equation variables include: melt driven unloading (q_{und}^{melt} , mm hr⁻¹) and canopy snow melt rate (q_{melt} , mm hr⁻¹), with the remaining variables defined in Table 2.

	Dimensionless Snowmelt Rate, Shear Stress	Dimensionless Snowmelt Rate	Air Temperature	Ice-Bulb Temperature
$p(a)$	$p < 0.05$	$p < 0.05$	$p < 0.05$	$p < 0.05$
b	$1.54 \times 10^{+01}$	NA	NA	NA
$p(b)$	n.s.	NA	NA	NA

A limitation of the melt driven unloading analysis presented in Table 3 is that canopy snow drip and unloading were both measured by the subcanopy lysimeters. To partition melting canopy snow into solid snow unloading and drip, five warm & humid events were selected, in which the median air temperature was above 0°C and relative humidity was above 65%, resulting in less than 5% contribution of dry snow unloading and sublimation to ablation as determined by the CP25 model. These periods differed from the aforementioned melt events as the weighed tree was used for this analysis which had different periods where this instrument was functioning properly. For these events, unloading was calculated using Equation 11 and canopy snowmelt rates were calculated using CRHM with Equation 3. The unloading-to-melt ratio over these events were found to be positively correlated with canopy snow load (Figure 4).

When canopy snow loads remained between 0 and 5 mm, the unloading-to-melt ratio varied from approximately 0 to 0.5. As snow loads increased, this ratio increased linearly, reaching its peak value of 5.0 for a canopy snow load of 30 mm (Figure 4). This relationship can be expressed through a linear function:

$$q_{und}^{melt} = R \cdot q_{melt}(L) \quad (13)$$

where R represents the unloading-to-snowmelt ratio and q_{melt} is the canopy snowmelt rate (mm hr⁻¹). Equation 13 is similar to Equation 33 in Andreadis et al. (2009); however, instead of a constant value of 0.4 for R , it was determined as:

$$R = m \cdot L + b \quad (14)$$

where m and b were determined as 0.15 and -0.43, respectively, using the generalised least squares regression shown in Figure 4 ($R^2 = 0.73$, RMSE = 0.79 (-)). The ordinarily least squares regression exhibited heteroscedasticity, with residual variance increasing at higher unloading-to-melt ratios (Breusch-Pagan test $p < 0.05$; see Supporting Information) and was accounted for by using a generalised least squares regression. See the Supporting Information section for additional statistical details. Additional observations of canopy snowmelt from the subcanopy rain gauges were also used to estimate R (Figure 4). The number of usable observations was limited to three events (out of the 5 warm & humid events) due to freeze-thaw events that seized up the tipping bucket mechanisms in the rain gauges, however, these measurements are still useful in providing some validation of the CRHM canopy snowmelt/drip calculations (Figure 5). The CRHM-estimated cumulative drip was higher than the subcanopy rain gauges for two out of the three melt events. Differences in the timing and magnitude of the observed and simulated values were expected due to both instrument uncertainties in the rain gauges from freezing of rain gauges and thawing of snow in the collection funnels, and in the canopy snow energy balance simulation.

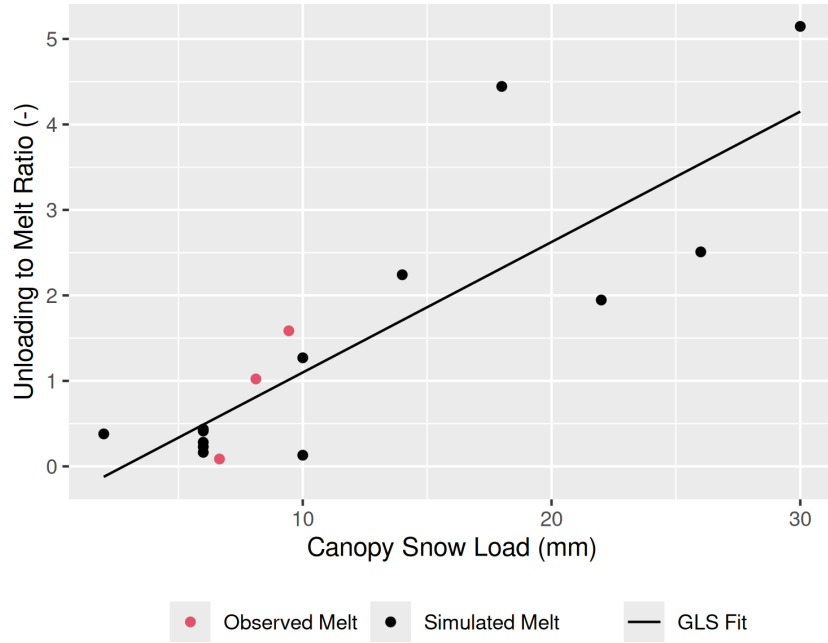


Figure 4: The ratio of canopy snow unloading (weighed tree residual) to snowmelt across different canopy snow load bins for each of the five warm & humid events. Black dots represent the observed cumulative unloading divided by the cumulative snowmelt simulated using the new canopy snow model (CP25). Red dots show the cumulative observed unloading divided by snowmelt measured by the rain gauges. Multiple dots within a bin correspond to different events.

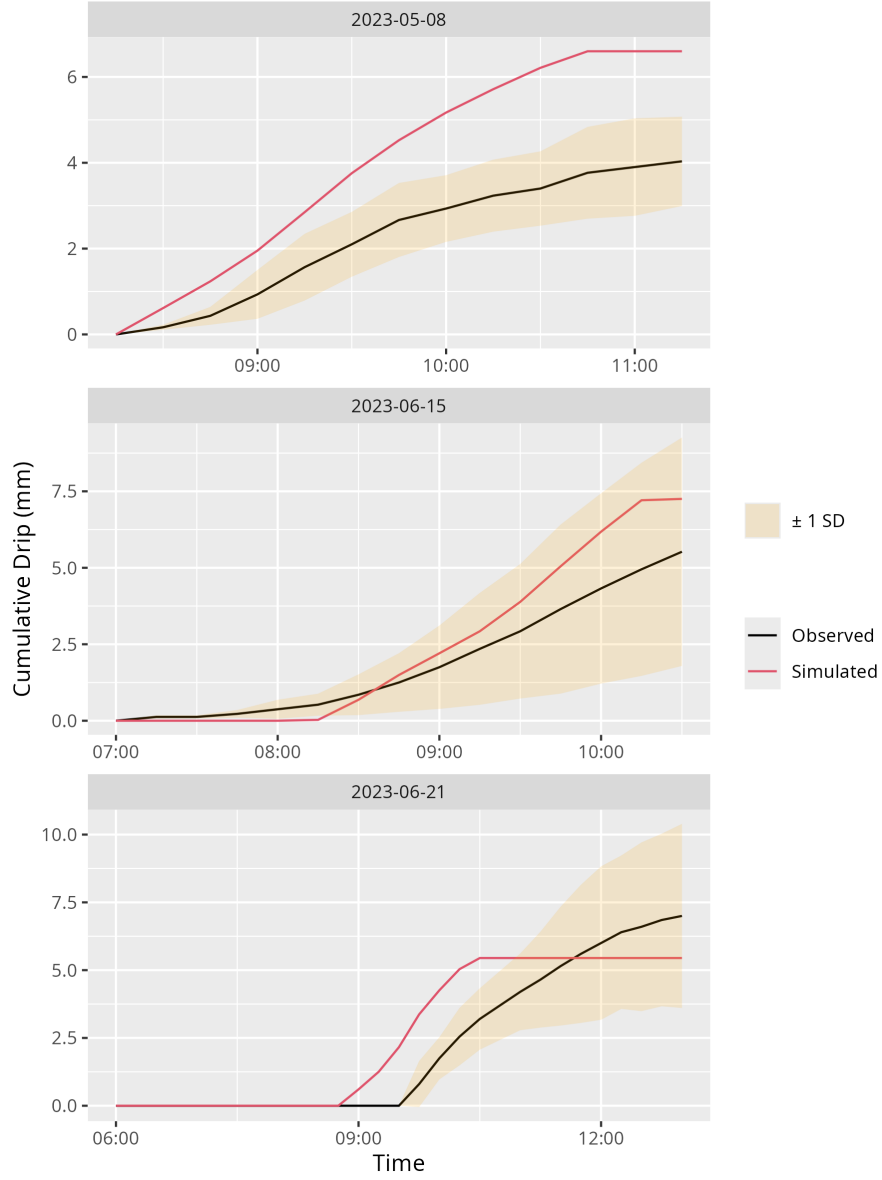


Figure 5: Cumulative canopy snow drip measured as the average of four subcanopy tipping bucket rain gauges (Observed) and simulated using the new canopy snow model (CP25) for three warm and humid events, with dates labelled as year-month-day.

3.2 New Canopy Snow Model

The CP25 model is based on the canopy snow mass balance formulation (Equation 1), where q_{tf} was represented as:

$$q_{tf} = [(1 - C_p)\alpha]q_{sf} \quad (15)$$

and C_p is the leaf contact area calculated using Equation 10 in Cebulski & Pomeroy (2025b) and α is an efficiency constant.

Since q_{unld}^{melt} naturally only occurs while intercepted snow is melting, and wind-induced unloading was also found to occur over the melt periods, total canopy snow unloading, q_{unld} (mm hr⁻¹), was calculated as:

$$q_{unld} = q_{unld}^{melt} + q_{unld}^{dry} \quad (16)$$

where melt-driven unloading (q_{unld}^{melt}) was modelled using Equation 13, while dry snow unloading (q_{unld}^{dry}) was represented using Equation 12. Canopy snow drip (q_{drip}) is derived from calculations of canopy snowmelt from Equation 3, with storage limited by the canopy liquid water holding capacity computed from Equation 2; any excess was assumed to immediately drain. Wind transport of canopy snow (q_{wind}^{veg}) is incorporated in the Equation 12 calculation. Sublimation of intercepted snow (q_{veg}^{sub}) was represented using Equation 10.

3.3 Event-based Evaluation of Canopy Snow Ablation Models

The updated canopy snow model (CP25), as well as the existing models R01, SA09, and E10 (Table 1) were evaluated using weighed tree lysimeter measurements of canopy load during seventeen canopy snow ablation events. These events had air temperatures ranging from -30.5°C to +6.9°C and wind speeds from calm to 5.3 m s⁻¹ (Figure 6). This range of meteorological conditions is particularly wide and spans conditions typical of the cold boreal to temperate maritime needleleaf forests. Events were classified as cold & dry, cold & humid, warm & dry, and warm & humid based on the median event air temperature (above or below 0°C) and relative humidity (above or below 65%) (Figure 6).

Simulated canopy snow load by the CP25 model closely matched the observations for all 17 events, demonstrating the most consistent agreement amongst the models evaluated (Figure 7). The large declines in canopy snow load for E10 (Figure 7) are due to the maximum canopy snow load used in this model which ranged from 7 to 12

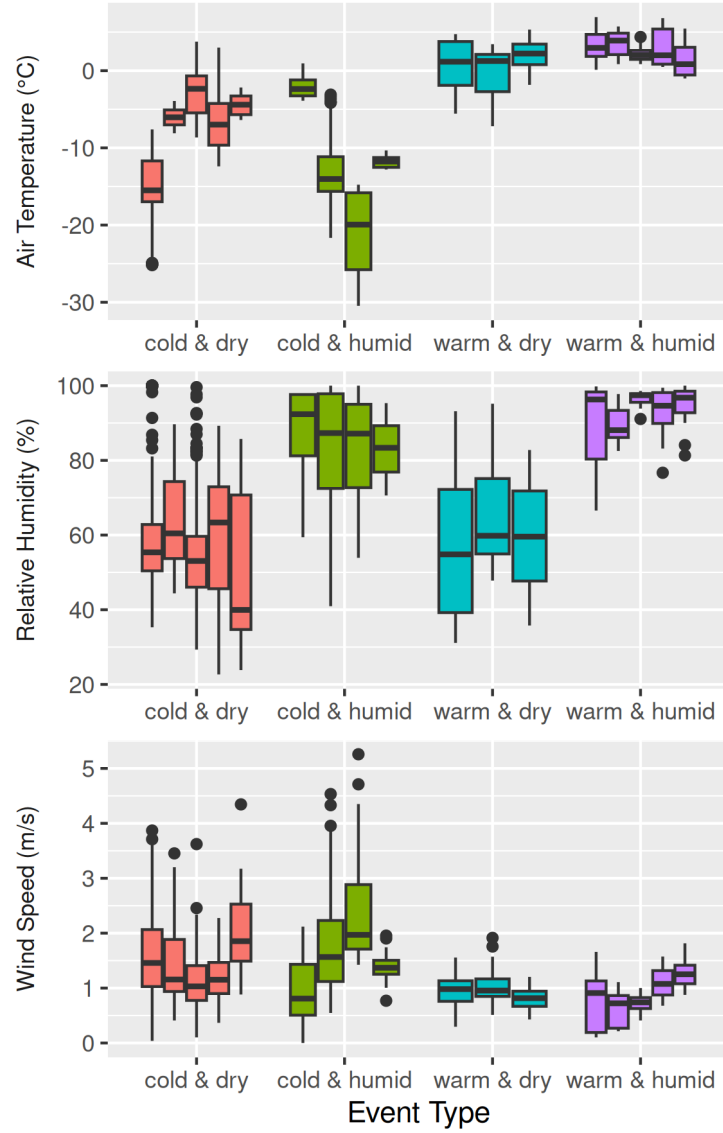


Figure 6: Boxplots of showing the distribution of meteorological measurements of air temperature, relative humidity, and wind speed over each of the seventeen selected ablation events. Note: the rectangle vertical extent represents the interquartile range (25th to 75th percentile), the horizontal line within each box indicates the median, and the whiskers extend to 1.5 times the interquartile range. Circular points beyond the whiskers represent outliers.

mm depending on the fresh snow density (a function of air temperature in Hedstrom & Pomeroy, 1998).

The energy balance-based snowmelt modelling approaches (CP25 & SA09) yielded the most accurate representation of canopy snowmelt over the warm & humid events. The CP25 mean bias of 0.02 mm hr^{-1} , which was smaller than the -0.11 mm hr^{-1} bias associated with SA09 (Figure 8). The improvement for CP25 over SA09 comes from its representation of the increase in unloading at higher canopy snow loads (Figure 4), as observed for the 2022-06-14 event (Figure 7). The air temperature (R01) and ice-bulb temperature (E10) models produced a wider range of mean biases (Figure 8) and larger mean bias of over 0.42 mm hr^{-1} during the warm & humid events, compared to CP25 and SA09. The rate of ablation was slower for canopy snow loads below $\sim 1.5 \text{ mm}$ and $\sim 0.3 \text{ mm}$ for CP25 and SA09 respectively; due to their differing liquid water storage capacities and unloading rates (Figure 7). For the warm events other than 2022-04-23, the observed decline in ablation rate occurs around 2 to 3 mm, exceeding the threshold predicted by all models.

The warm & dry events had consistent performance with mean biases of 0.02 mm hr^{-1} for each of the four models (Figure 8). Initiation of ablation was delayed compared to observations from the weighed tree for the CP25 and SA09 models for all three of the warm & dry events (Figure 7). The temperature threshold methods (E10 & R01) achieved better timing on the onset of ablation for two events (2022-03-29 and 2022-04-21) compared to CP25. However, E10 initiated the onset of ablation slightly earlier for 2022-04-23 and the rate of ablation was also lower than observed for this event.

For the cold & dry events, all models had accurate performance with mean biases ranging from -0.01 to 0.01 mm hr^{-1} , with a slight improvement in the mean bias for CP25 of $-0.008 \text{ mm hr}^{-1}$ (Figure 8). Canopy snowmelt was overestimated by the E10 model and caused the steep initial decline in canopy snow load on 2022-03-02 which is not registered by the weighed tree or other models. However, underestimates of ablation compensated for this overestimate over the remaining cold & dry events—which had moderate wind speeds—as wind-driven unloading is not included in the E10 model (Figure 6). For events 2022-03-02 and 2023-03-14 the R01 model overestimated canopy snow ablation due to an overestimation of wind-driven unloading (Figure 7).

The importance of representing wind-driven unloading was clear during the cold & humid events, where the mean bias of models including this mechanism was reduced compared to other approaches; for example, 0.04 mm hr^{-1} for R01 and 0.17 mm hr^{-1} for CP25. In contrast, simulations that did not explicitly account for wind-driven

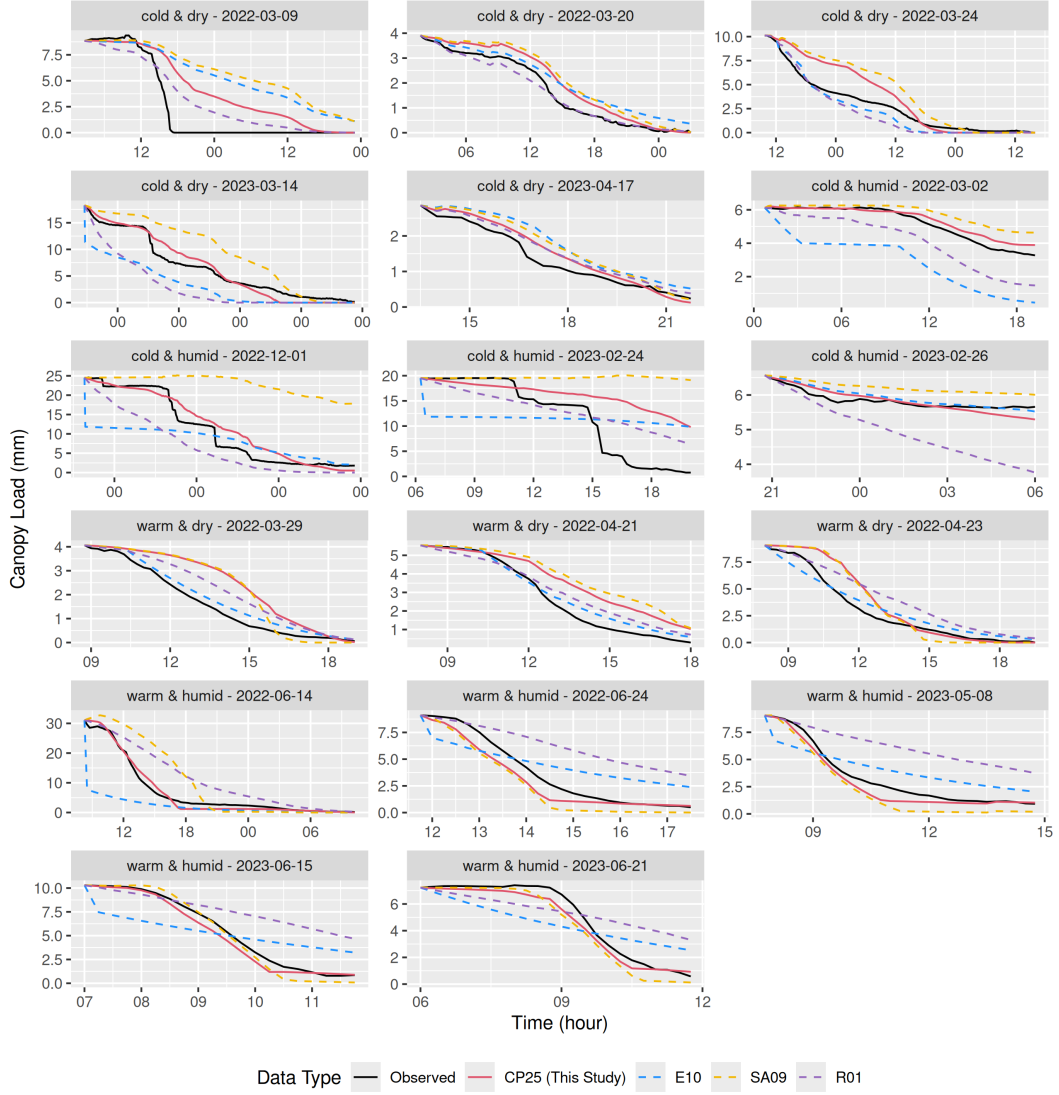


Figure 7: Time series of canopy snow load for individual events measured by the weighed tree (observed) and simulated using the four canopy snow models, with dates labelled as year-month-day.

unloading exhibited higher biases, exceeding 0.32 mm hr^{-1} (Figure 8). Although the E10 model does not include wind-driven unloading, it performed best for the 2023-02-26 event due to its relatively slow time-based unloading rate compared to CP25 and R01 which overestimate ablation for this event (Figure 7). The R01 model overestimated unloading over most of the cold & humid events and had a higher median bias for the cold & humid events compared to CP25 (Figure 8). The CP25 model had consistently lower bias across the three cold & humid events, but still underestimated ablation for the 2023-02-24 event which had peak wind speeds of over 5 m s^{-1} . Over this event, 1.3 mm of snow was measured at a shielded precipitation gauge in a nearby clearing and was likely derived from wind transport of snow from the canopy, as clear skies with no precipitation were observed. The amount of snow observed to unload from the canopy into the subcanopy lysimeters during this event was consistent with simulated unloading in CRHM suggesting that the remaining unaccounted-for snow was likely entrained into the atmosphere and sublimated and/or transported to distant sites.

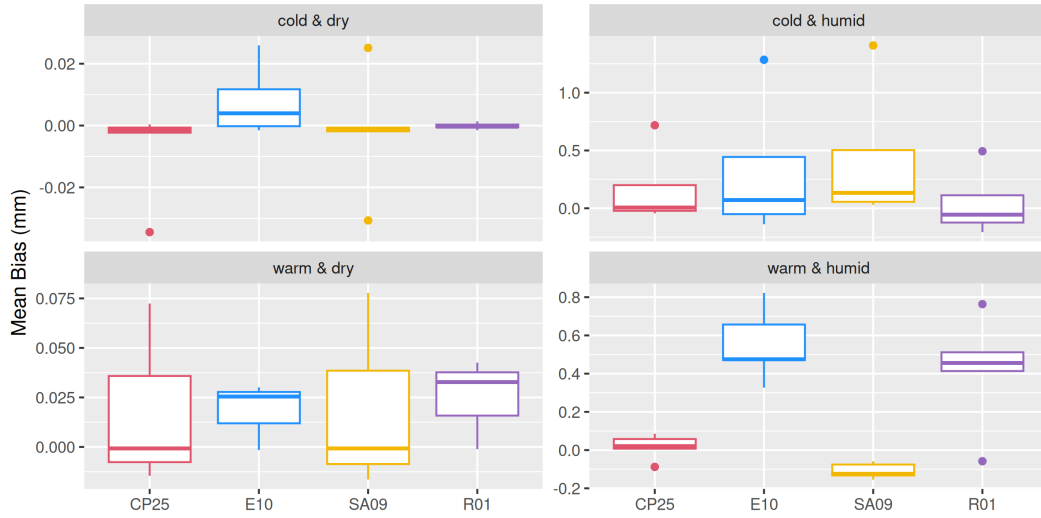


Figure 8: Boxplots illustrating the distribution of event mean biases in canopy snow ablation calculated between simulations and observations from the weighed tree. The vertical extent of each rectangle represents the interquartile range (25th to 75th percentile), the horizontal line within each box indicates the median, and the whiskers extend to 1.5 times the interquartile range. Circular points beyond the whiskers represent outliers. The diamonds represent the mean of the event biases.

3.4 Bootstrap Resampling Evaluation of Canopy Snow Ablation Models

Based on the bootstrap analysis, which evaluated model performance across resampled combinations of canopy snow ablation events, the CP25 model consistently demon-

strated the highest performance across all error metrics, including mean bias, RMSE, NSE, and KGE (Figure 8). In contrast, the E10 and SA09 models showed the poorest performance, due to the absence of key processes such as wind-induced unloading (E10 and SA09) and energy-balance-based melt (E10 and R01). The R01 model showed the second-highest performance, which may reflect the inclusion of both melt and wind-induced unloading processes, although its reliance on an air-temperature index melt formulation likely explains its reduced accuracy relative to CP25.

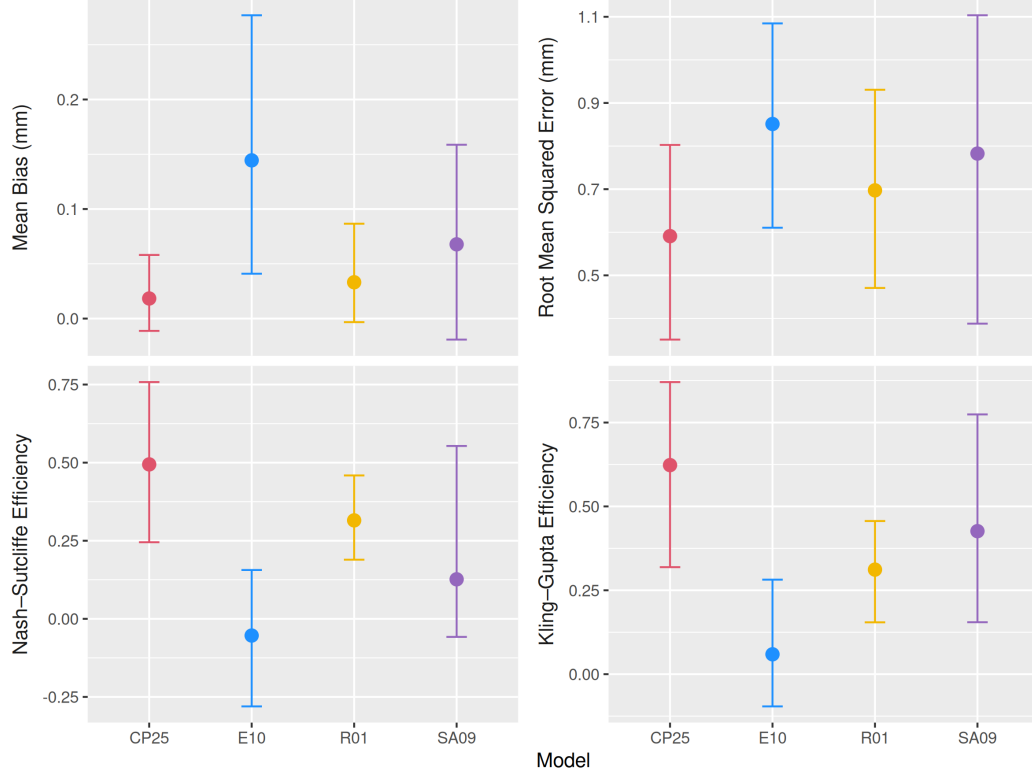


Figure 9: Error statistics from bootstrap resampling differing combinations of canopy snow ablation events (1000 replicates). Points indicate the mean metric and error bars show the 95% confidence intervals estimated across all resampled event combinations.

3.5 Canopy Snow Partitioning

During warm & humid events, all four parameterisations showed relatively consistent partitioning of canopy snow, with only a small fraction returned to the atmosphere (Figure 10). The warm & dry events had greater variability in the partitioning of intercepted snow—when compared to the warm & humid events—with a larger contribution from sublimation and evaporation processes. Increased unloading from the E10 model resulted in a greater fraction of intercepted snow reaching the ground compared to the CP25 model over the warm & dry events.

For the cold & dry and cold & humid events, the two parameterisations that include wind-driven unloading (R01 and CP25) had differing fractions of snow partitioned to the ground via unloading. A higher fraction of intercepted snow reached the ground for R01 due to the higher dry snow unloading rate compared to CP25 over all of the cold events. For both the cold & dry and cold & humid events SA09 partitioned all snow back to the atmosphere via sublimation as dry snow unloading is not included in this parameterisation. Although CP25 and E10 include differing unloading processes, when averaged over all events, both had a similar fraction of snow reaching the ground (70%) versus the atmosphere (30%). SA09 has the largest discrepancy which returned 40% of intercepted snow back to the atmosphere and the R01 with the least amount of snow reaching the atmosphere with 24%.

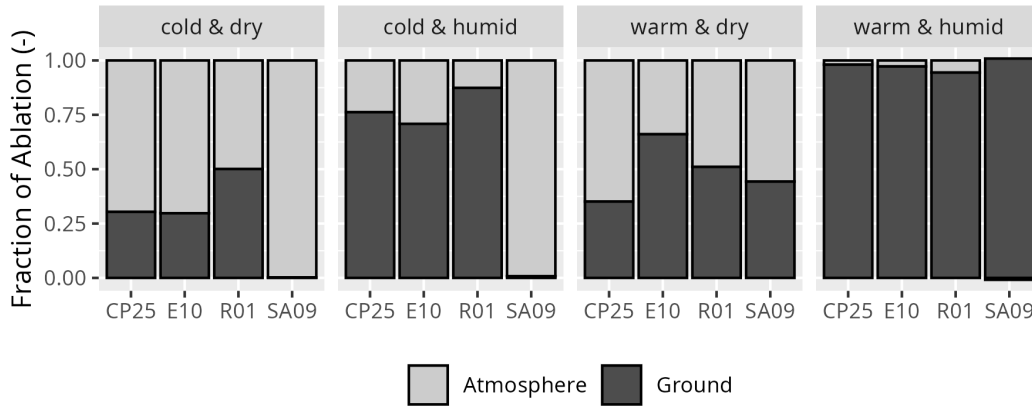


Figure 10: Bar chart illustrating the proportion of intercepted snow that was either lost to the atmosphere as sublimation and/or evaporation of melted snow or transferred to the ground through unloading or drip of melted snow by each event type for all 17 events.

4 Discussion

4.1 Processes Governing Canopy Snow Unloading

Observations of canopy snow unloading support the hypothesis that unloading is primarily controlled by snowmelt and dry-snow related processes, both of which are also influenced by the amount of snow intercepted in the canopy. The ratio of unloading to canopy snowmelt was found to increase linearly with increasing canopy snow load (Figure 4), which differs from Storck et al. (2002) who originally found the ratio of canopy snow unloading to melt to be constant. The measurement difficulties noted by Storck et al. (2002) limited their estimate of this ratio to a single mid-December event, preventing any association with canopy snow load. Previous studies have identified relationships between melt-induced unloading and various meteorological

parameters, including empirical functions of air temperature (Katsushima et al., 2023; Roesch et al., 2001), ice-bulb temperature (Ellis et al., 2010; Floyd, 2012), and solar radiation (Katsushima et al., 2023). Although branch bending and subsequent unloading has been shown to be associated with air temperature (Schmidt & Gluns, 1991; Schmidt & Pomeroy, 1990), observations here indicate that temperature-related unloading increases occur primarily near the freezing point, eliminating the need for a separate temperature parameterisation beyond snowmelt-associated unloading processes. Furthermore, the statistical significance of shear stress as a predictor of unloading during melt events (Table 3), supports the application of the wind-induced unloading parameterisation during canopy snowmelt periods. However, further research is required to evaluate the efficiency of wind-driven unloading as snow transitions from dry to melting.

Dry snow unloading was found to increase exponentially with wind speed and linearly with shear stress (Table 2). These results differ from earlier research that has represented this process as a linear function of wind speed and canopy load (Bartlett & Verseghy, 2015; Katsushima et al., 2023; Roesch et al., 2001), as shown in the Supporting Information. The slightly higher performance of the shear stress relationship compared to wind speed—when excluding melt events—is likely due to the physical relationship between shear stress force and kinetic energy transfer to the canopy, wind transport from the canopy, and movement of branches in the canopy induced by drag (shear) forces. The differing relationship presented here, compared to the R01 model, may be attributed to the development of the R01 parameterisation using above canopy albedo as a proxy for canopy snow unloading (Bartlett & Verseghy, 2015; Roesch et al., 2001). This approach would have included both unloading and sublimation processes, in addition to greater measurement uncertainties (Cebulski & Pomeroy, 2025a), and may have contributed to R01 overestimating wind-driven unloading (Figure 8). Conversely, the subcanopy lysimeter measurements employed here provided a more direct quantification of canopy snow unloading rates. Additional factors that are not considered in the new parameterisations include wind erosion, branch movement, structural degradation, bond weakening, increased elasticity of branches (associated with species and age), snow density, and liquid water content. These factors would all directly have implications for the canopy snow ablation routines presented in this study. For instance, variation in branch flexibility amongst tree species and age classes can alter how snow load affects branch angle, thereby changing the likelihood of unloading. The addition of liquid water content in the canopy snow due to phase change can increase (Pomeroy & Gray, 1995) or decrease , Figure 4 cohesion and adhesion of snow clumps within the canopy which may also

affect the dry-snow unloading rate as meltwater refreezes, although this effect was not observed in this study. Sublimation of intercepted snow reduces its structural integrity (MacDonald, 2010), and equi-temperature metamorphism may increase density and cohesion both with differing associations with canopy snow unloading rates and the exposure of intercepted snow to turbulent energy exchanges, although these factors did not emerge as dominant processes in this study.

4.2 Performance Comparison of Ablation Models

The improved performance of the new CP25 model across a wide range of meteorological conditions demonstrates the advantage of incorporating both dry snow and melt driven unloading processes and physically-based energy balance to simulate ablation (Figure 9). In contrast, existing models were limited by missing processes such as wind-driven unloading (SA09 and E10) or relied on temperature-dependent parameterisations of melt and drip processes which had limited transferability across differing events (R01 and E10). The constant canopy snow unloading to melt ratio of 0.4 in the SA09 model led to reduced performance for one melt event with canopy snow loads up to 30 mm and for all melt events for snow loads less than 1 mm, whereas the CP25 model more accurately predicted ablation a range of snow loads during the melt events (Figure 7). The low liquid water retention capacity and increased solid snow unloading for lower snow loads also contributed to an underestimation of canopy loads during the end of most melt events for SA09 (Figure 7).

Models utilising temperature-based canopy snowmelt parameterisations (E10 and R01) exhibited inconsistent performance, particularly during warm & humid events where they underestimated ablation (Figure 8). This was due to their reliance on air temperature or ice-bulb temperature as proxies for energy input into the canopy, which failed to represent available energy over these events. In contrast, all four models performed similarly during warm & dry events, where air temperatures were closer to 0°C, and lower humidities increased the impact of cooling from canopy snow sublimation. The superior performance of the energy balance-based canopy snowmelt models (CP25 and SA09) during warm & humid events likely reflects their ability to represent the differing energy balance over these events. The maximum canopy snow load threshold implemented in the E10 model was lower than observations from the weighed tree. This limitation offset its tendency to underestimate unloading processes as was also observed by Lundquist et al. (2021) and Lumbrazo et al. (2022). In this study, the CP25, SA09, and R01 models did not include a maximum snow load, and their performance aligns with the hypothesis of Lundquist et al. (2021) that this limit

may be unnecessary—or much higher than previously thought—when combined with a comprehensive canopy snow ablation routine.

CP25 provided a better representation of wind-driven unloading compared to R01 (Figure 8), substantial errors persisted across all models during these events. Wind unloading parameters also may be influenced by tree species, forest structure, and snow characteristics (Lumbrazo et al., 2022), necessitating further field-based research on canopy snow unloading in diverse environments to assess their broader applicability. All models underestimated ablation for the strongest wind-dominated event (2023-02-24, Figure 7). Wind transport of canopy snow to the nearby Powerline snowfall gauge occurred during this event, but was a small fraction of canopy snow ablation, 1.3 mm of a total of 20 mm of canopy snow ablation. Since unloading measured by the subcanopy lysimeters corresponded well with CP25 predictions for this event, the approximately 9 mm of unaccounted canopy snow ablation may be attributed to uncertainties within the wind-driven unloading parameterisation (Figure 3) and possible atmospheric entrainment of canopy snow that was either transported to distant locations and/or sublimated. These findings align with observations by Troendle (1983) but contrast with Hoover & Leaf (1967), who proposed that most wind-transported snow relocates to nearby sites with minimal sublimation effects.

4.3 Canopy Snow Partitioning

Substantial variability was found in the fraction of snow that sublimated and/or evaporated as liquid meltwater versus unloading and drip depending on the canopy snow ablation model selected (Figure 10). For example, the exclusion of wind-driven unloading processes in the SA09 model resulted in 100% of intercepted snow reaching the atmosphere via canopy snow sublimation for both the cold & dry and cold & humid events. This differed considerably from the CP25 model which returned 70% and 24% of snow back to the atmosphere for the cold & dry and cold & humid events, respectively. Although E10 and CP25 include differing process representations, they predicted comparable fractions of snow reaching the ground versus returning to the atmosphere (Figure 10). The agreement between CP25 and E10 is notable, since the E10 has been tested and found to perform well in predicting subcanopy snowpacks around the world (Ellis et al., 2010; Gelfan et al., 2004; Pomeroy et al., 2022; Sanmiguel-Vallelado et al., 2022). However, the results shown here reveal that E10's individual process representations can be in error, particularly under warm and windy conditions, potentially explaining the difficulties when applying E10 at locations where parameterisation errors fail to offset one another (Lumbrazo et al., 2022; Lundquist et al., 2021). For example, at locations which intercept a larger

amount of snow, the E10 maximum canopy snow load would overestimate the amount of unloading, and a greater deviation between the E10 and CP25 model is expected.

4.4 Future Directions

Physically based approaches such as CP25 are particularly relevant for predictions of snowfall regimes in forested basins under a changing climate, where warming may reduce the reliability of empirically derived canopy snowmelt models like E10 and R01. Other regions that are transitioning from cold dry snow unloading and canopy snow sublimation dominated regimes to increased drip and melt-driven unloading processes with climate warming (e.g., Fang & Pomeroy, 2020) are expected to also benefit from this comprehensive representation of canopy snow ablation processes. Although the revised model performed best for this dataset, further validation is required across a wider range of climates and forest compositions.

The density of snow intercepted in the canopy is expected to influence both dry snow and melt-induced unloading processes—and is incorporated in the E10 parameterisation for the initial accumulation component based on the findings of Schmidt & Gluns (1991)—but is not explicitly represented in any of the ablation calculations included in this study. Fresh, low-density snow typically exhibits lower cohesion and adhesion compared to older snow, which may have undergone freeze-thaw cycles or equi-temperature metamorphism, processes that increase snow density and bond strength hence, increasing the mechanical resistance to unloading (Pomeroy & Gray, 1995). While vapour deposition and rime-ice accumulation are simulated in some models [e.g., Clark et al. (2015); Ellis et al. (2010); this study] via the latent heat flux parameterisation, it is usually treated as additions to the canopy snow reservoir where the unloading rate is typically overestimated (Lumbrazo et al., 2022). Due to the cold continental location of this study, canopy rime and/or frost deposition was not observed, and thus the unloading rates presented here would also likely overestimate unloading rates for ice. Thus, the transferability of these results may be limited in maritime climates where deposition of rime and frost is more likely due to the higher humidity and warmer temperatures. The addition of a canopy ice reservoir could help better represent this processes, however further research is needed to investigate how ablation rates differ in these environments.

This study implemented a single layer snow model for snow intercepted in the canopy. The unloading relationships developed here, along with the parameterisation for initial interception presented in Cebulski & Pomeroy (2025b), are expected to be applicable in multi-layer canopy snow models (e.g., Clark et al., 2015; Essery et al., 2025) with modifications to the energy balance calculations presented in Section 2.5.

However, uncertainties in generating forcing data and collecting canopy snow ablation measurements at different canopy heights will likely influence the performance of these relationships when applied vertically within the canopy.

Key limitations remain in measuring canopy snow sublimation using eddy correlation systems (Conway et al., 2018; Harding & Pomeroy, 1996; Harvey et al., 2025; Helgason & Pomeroy, 2012b; Parviainen & Pomeroy, 2000) and separating snow unloading from meltwater drip (Floyd, 2012; Storck et al., 2002), which limit the development and testing of canopy snow ablation parameterisations. Since unloading, melt, and sublimation are competitive ablation processes, correctly representing each of them is essential for determining the fraction and phase of snow reaching the ground, and the duration that the canopy is snow covered, an important factor for land-surface albedo calculations (Cai et al., 2025; Henderson et al., 2018; Thackeray et al., 2014). Whilst separating initial interception and ablation processes (Cebulski & Pomeroy, 2025a, 2025b), will improve process representations, these routines still need to be evaluated together against additional field observations across different sites, climates, forest types, and spatial scales to assess model transferability and performance.

5 Conclusions

Canopy snow ablation processes govern the partitioning of intercepted snowfall, yet their representation in modelling frameworks remains uncertain due to insufficient process-level validation. This study represents the first development and validation of an unloading model addressing both energy balance-based melt and dry snow unloading processes together. Novel parameterisations for dry snow and melt-induced unloading were introduced, with key differences from previously established approaches. Canopy snow load and shear stress were found to be the best predictors of dry snow unloading ($R^2 = 0.58$). The canopy melt rate exerted the strongest control on snow unloading during melt events ($R^2 = 0.73$), consistent with one existing model, and show improved performance compared to temperature index melt routines. A new finding was that the ratio of unloading to canopy snowmelt increased with canopy snow load. In addition to this empirical evidence, physical processes such as structural degradation, snow particle bond weakening, lubrication of wet canopy snow during melt, and the shear force exerted on canopy snow by wind further support representing these processes. Moreover, an existing approach which used the concept of a maximum intercepted snow load greatly underestimated the canopy snow storage capacity when compared to observed snow loads from weighed tree measurements. Throughout the two-years of observations presented here, a maximum canopy snow load was not observed, likely as unloading rates increased with higher snow loads.

Wind transport events were relatively rare in this wind-exposed subalpine forest, but resulted in a considerable underestimation of the amount of snow returned to the atmosphere or surrounding sites during one event.

A new canopy snow ablation model that incorporates an updated canopy snow mass and energy balance demonstrated improved accuracy ($\text{NSE} = 0.49$; $\text{KGE} = 0.62$) compared to existing approaches ($\text{NSE} = -0.05$ – 0.32 ; $\text{KGE} = 0.06$ – 0.43) across varied meteorological conditions. Existing models failed to maintain accuracy across events with a wide range of meteorology due to neglect of key processes such as wind-driven unloading and/or empirical representations of melt processes. The greatest inter-model discrepancies in canopy snow load occurred during warm and humid events, where temperature-based canopy snowmelt parameterisations showed substantially higher mean biases relative to energy balance-based models.

Amongst the models tested, the largest errors were found during cold, humid wind-driven unloading events, though performance was improved when incorporating a site-specific shear stress-based parameterisation. Partitioning of intercepted snow disposition between the ground and atmosphere varied most amongst cold events, where neglecting the dry snow unloading process resulted in considerable overestimates of canopy snow sublimation losses. Observations were collected over a wide range of meteorological conditions typical of cold boreal to temperate maritime climates, with air temperatures from -30.5°C to $+6.7^{\circ}\text{C}$ and wind speeds from calm to 5.3 m s^{-1} . Some processes, such as rime-ice and frost deposition, were not observed. The study was conducted in a subalpine forest of mature fir and spruce, so unloading rates and process emergence may differ in forests with other species and ages due to differences in branch flexibility, structure, and impacts on the energy balance. Incorporating the updated unloading schemes developed here could improve the representation of canopy snow ablation and, by extension, the partitioning of precipitation and canopy albedo in hydrological and land surface models. Nonetheless, further testing is needed across various climate and forest compositions to assess model transferability.

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856 **7 Data & Software Availability Statement**

857 The Cold Regions Hydrological Model Platform (CRHM) source code is available at
858 <https://github.com/srlabUsask/crhmcde>. Model forcing data, model outputs, valida-
859 tion data, processed data, and scripts are available at [https://doi.org/10.5281/zen-](https://doi.org/10.5281/zenodo.16898881)
860 [odo.16898881](https://doi.org/10.5281/zenodo.16898881).

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